

TUMNIN, Russian Federation

WMO: 316830

Lat: 49.6708N

Lon: 140.1013E

Elev: 189

StdP: 14.60

Time Zone: 10.00 (E10)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.3	-27.4	-35.6	0.7	-30.6	-31.6	0.9	-26.9	15.4	8.0	11.2	11.4	0.1	0	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.4	84.0	67.9	80.2	66.7	76.7	64.9	70.8	80.3	68.7	77.2	66.7	74.1	4.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.4	101.5	75.0	65.6	95.2	73.1	63.8	89.2	70.9	34.9	80.5	33.1	77.6	31.5	74.2	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	11.8	9.7	7.8	DB	-37.7	93.1	4.3	4.2	-40.8	96.1	-43.3	98.5	-45.7	100.9	-48.8	103.9
(5)				WB	-37.8	76.2	4.1	4.4	-40.7	79.3	-43.1	81.9	-45.3	84.3	-48.3	87.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	33.2	-4.1	3.5	18.5	35.2	47.3	57.5	64.4	64.6	54.3	38.3	17.1	0.2	(6)
(7)		DBStd	25.66	11.51	9.56	9.29	6.48	6.68	6.99	5.41	5.05	6.52	6.89	11.17	11.17	(7)
(8)		HDD50	7474	1676	1303	977	445	133	11	0	32	366	988	1543		(8)
(9)		HDD65	11740	2141	1723	1442	894	550	243	78	71	325	827	1438	2008	(9)
(10)		CDD50	1347	0	0	0	1	49	235	446	452	160	4	0	0	(10)
(11)		CDD65	137	0	0	0	0	1	17	58	58	3	0	0	0	(11)
(12)		CDH74	1391	0	0	0	1	53	363	570	365	39	0	0	0	(12)
(13)		CDH80	357	0	0	0	0	15	112	163	62	5	0	0	0	(13)
(14)	Wind	WSAvg	2.4	1.9	2.3	2.6	3.0	3.2	2.5	2.3	1.9	2.1	2.7	2.4	2.2	(14)
(15)	Precipitation	PrecAvg	24.7	0.8	0.7	0.8	1.7	2.7	2.4	3.3	3.8	3.5	2.2	1.3	1.3	(15)
(16)		PrecMax	32.6	2.4	3.8	2.2	3.2	5.1	6.1	8.0	8.7	7.9	5.6	2.8	4.5	(16)
(17)		PrecMin	15.5	0.0	0.0	0.2	0.2	0.6	0.7	0.2	0.6	1.3	0.3	0.1	0.1	(17)
(18)		PrecStd	3.8	0.6	0.9	0.5	0.8	1.2	1.1	2.2	1.9	1.9	1.5	0.9	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.1	34.4	48.5	67.9	81.0	89.8	90.4	85.7	78.4	66.1	44.9	31.4	(19)
(20)		MCWB	27.0	27.7	38.1	50.0	58.0	66.8	70.5	71.3	63.8	53.0	37.4	29.8	(20)	
(21)		2%	DB	25.0	28.1	41.2	58.7	73.0	82.5	84.7	81.4	72.9	59.1	40.2	27.8	(21)
(22)		MCWB	22.5	24.2	33.3	45.1	54.9	66.0	68.9	68.7	61.6	48.2	35.7	25.7	(22)	
(23)		5%	DB	20.8	23.8	37.3	52.4	67.2	77.4	80.4	78.2	68.7	54.2	36.6	23.4	(23)
(24)		MCWB	18.5	20.6	31.0	41.3	53.0	63.8	67.3	67.2	59.6	45.9	32.7	21.6	(24)	
(25)		10%	DB	15.4	19.7	33.5	47.5	62.0	72.9	76.3	74.8	65.0	50.2	33.5	18.7	(25)
(26)		MCWB	13.1	17.0	28.8	38.4	50.3	61.0	65.6	65.9	57.9	44.0	30.1	16.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.6	28.6	38.8	51.3	61.1	71.1	73.4	73.6	67.1	54.2	39.6	30.6	(27)
(28)		MCDWB	29.0	33.0	47.5	66.6	77.0	85.6	83.8	83.7	74.4	61.2	43.3	31.2	(28)	
(29)		2%	WB	22.6	24.3	34.1	45.9	57.1	67.7	70.9	70.8	63.8	50.4	36.1	25.4	(29)
(30)		MCDWB	24.7	28.1	40.2	56.8	70.1	78.2	81.7	78.3	69.5	56.1	38.7	27.2	(30)	
(31)		5%	WB	18.5	20.9	31.5	42.1	54.0	64.9	68.8	68.9	61.4	47.5	33.3	21.6	(31)
(32)		MCDWB	20.6	23.9	36.1	51.3	65.0	75.3	78.4	75.6	66.5	52.7	36.2	23.5	(32)	
(33)		10%	WB	12.9	17.0	29.3	39.1	51.2	62.1	66.6	66.9	59.1	44.6	30.3	16.5	(33)
(34)		MCDWB	15.2	19.8	33.5	46.5	61.0	71.8	74.7	72.8	64.0	49.4	33.2	18.5	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	25.2	28.6	25.9	20.7	23.3	24.1	20.2	18.4	21.0	22.1	20.2	21.9	(35)
(36)		MCDBR	17.6	29.3	29.8	32.9	35.7	34.9	29.9	25.3	27.1	30.5	19.4	16.7	(36)	
(37)		MCWBR	16.0	25.0	21.6	19.3	19.4	19.0	15.6	13.0	15.2	19.4	14.4	15.5	(37)	
(38)		5% WB	MCDBR	16.1	28.0	25.3	31.2	31.8	31.1	26.2	22.0	22.2	27.4	16.9	16.1	(38)
(39)		MCWBR	14.2	23.9	18.3	18.7	18.1	17.8	14.9	12.5	13.2	18.4	12.8	15.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.283	0.321	0.395	0.491	0.547	0.516	0.485	0.430	0.387	0.398	0.348	0.291	(40)	
(41)		taud	2.098	1.988	1.841	1.775	1.801	1.978	2.130	2.308	2.355	2.128	2.079	2.082	(41)	
(42)		Ebn at Noon	233	250	248	237	231	240	246	256	255	220	201	208	(42)	
(43)		Edh at Noon	26	38	53	64	65	55	47	37	32	34	27	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	464	829	1229	1412	1534	1661	1532	1348	1120	719	430	335	(44)	
(45)		RadStd	37	44	84	146	124	174	156	121	101	48	36	36	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.63	+2.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page